

The Great OS Migration

- The Environment:
 - Machines in our OSD and O3 infrastructure
- Catalyst for Change:
 - Because Leap 15.6 reached its EOL
- The Naive Assumption:
 - Just translate ifcfg files to nmconnection files, how hard could it be?
- Where this all started?
 - Upgrade a single o3 worker to openSUSE Leap 16.0

The multi-machine chaos

- Gre-tunnel fixes:
 - gre-tunnel script invocation with NM
 - NM arguments for dispatcher script - gre-tunnel
- Multi-machine setup:
 - OVS DB and NM states cleanup
 - br1 address conflict resolution - Failed attempt
 - br1 address conflict
 - Wrong approach for reloading Network State Changes
 - Fixed - Reload Network State Change

The salt sanity for OSD

- Gre-tunnel script generation:
 - gre-tunnel script normalization for NM
- New salt grain – network_backend:
 - Detect Network Stack via salt grain
- NetworkManager System Connections:
 - OSD Wokers bond interface for NM
- The D-BUS Culprit:
 - dbus max_connections_per_user limit
- Global IPv6 - the EUI-64 format:
 - EUI64 IPv6 for NM
- Limit python3-dataclass installation on old OS:
 - python3-dataclass installation

The “Gremlins” in the Network

- The Firewalld Asymmetric Routing Trap:
 - Issue: Firewalld’s IPv6 `rpfilter_DROP` wasn’t allowing SSH login
 - Workaround:
 - manually manipulated route-metric to prioritize VLAN, aligning the return paths and pacifying firewalld’s Reverse Path Filtering
 - The Fix:
 - [Disable Router Advertisement on Hypervisors](#)